MULTI-OMICS DATA ANALYSIS TO DISCOVER NOVEL BIOMARKERS FOR HEPATOCELLULAR CARCINOMA

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ABSTRACT

The severity of hepatocellular carcinoma (HCC) and lack of good diagnostic and prognostic markers make its management a significant challenge. Biomarkers like AFP, AFP-L3, DCP and staging systems such as BCLC and fibrosis score aid in assessing disease progression but lack precision in detecting early-stage HCC, hindering timely intervention and improved outcomes. In this study, we leveraged the TCGA-LIHC database to perform multi-omics analysis: a supervised DIABLO analysis integrated mRNA, miRNA and protein data to build a robust classification model, tuned for accurate, unbiased classification of tumor and non-tumor samples. G:Profiler was used for pathway enrichment analysis of the top 100 mRNAs (PC1 positive, PC2 negative). PC1 genes showed an enrichment for pathways involved in cell adhesion, immune response (p-value < 0.05), platelet degranulation and activation, and oncogenic MAPK signaling, suggesting primary source of variation within our samples is related to the tumor's interaction with its microenvironment and the host immune response. Similarly, PC2 genes revealed enrichment of pathways related to cell cycle and proliferation along with extracellular matrix organization and sodium and inorganic ion transmembrane transport indicating the secondary distinction between the tumor and non-tumor phenotypes is due to upregulation of processes controlling cell division and growth, which is consistent with tumorigenesis. These top components (mRNA) need to be further analysed as potential diagnostic biomarkers.

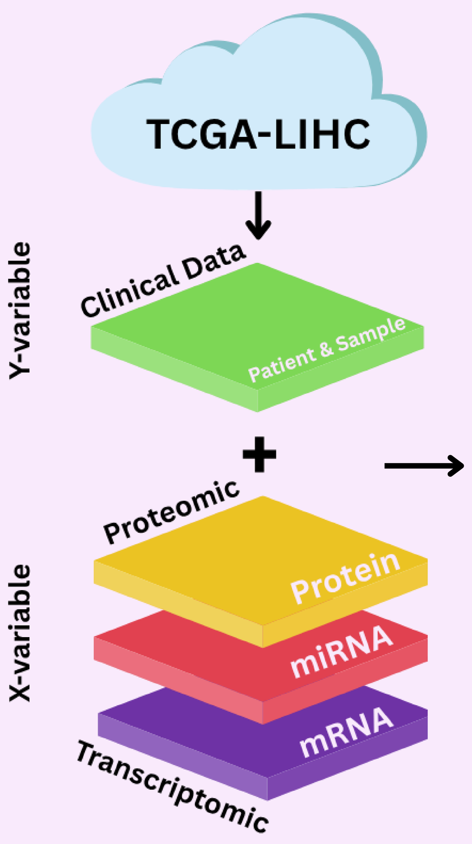
OBJECTIVE

Develop and validate a robust multi-omics classification model.

Identify and characterize feature by performing pathway enrichment analysis.

Evaluate the diagnostic potential of the identified features.

METHODOLOGY



Data_Final	Sample X Feature
mRNA	164 X 18908
miRNA	164 X 1324
Protein	164 X 219
Clinical	164 X 70

Filtered out near zero-variance and duplicate features

Ensuring common samples and consistent order

Handling Missing Data and NaNs values

Data	Sample X Features
mRNA	420 X 20530
miRNA	423 X 2172
Protein	184 X 219
Clinical	438 X 110

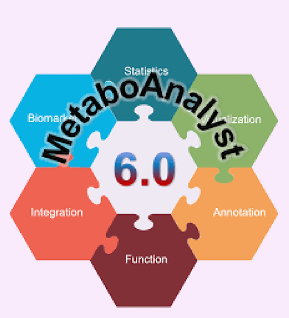
[1]

DIABLO

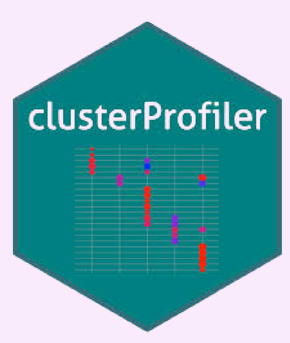
Data Integration Analysis for Biomarker discovery

Multi 'omic (>2) N-Integration

Input Data → Pre-Processing → Integration → Feature Selection → Pathway Analysis → Performance



BIOMARKER ANALYSIS [3]



Enriched GO:biological process for Gene Symbols in two components **(adjusted p_value <0.05)** [2]

Optimal Component selection → Tuning → Variable Contribution → Correlation between features

DISCUSSION

- **Synergistic Power of Multi-Omics Integration**: The integrated DIABLO model validates the hypothesis that combining heterogeneous omics data provides a more comprehensive and biologically relevant view of HCC than single-omics analysis alone.
- **Mechanistic Drivers of HCC Heterogeneity**: Pathway enrichment confirms that the two primary components distinguishing samples are driven by distinct biological forces: PC1 captures tumor-microenvironment and metabolic process (potential therapeutic targets), while PC2 highlights the fundamental dysregulation of cell division.
- **Clinical Potential:** The identified top-performing genes represent powerful candidates. These markers could be leveraged to build a highly sensitive diagnostic panel for early-stage HCC, a phase where current markers (like AFP) are lacking in precision.
- **Translational Potential for Early Diagnosis**: The good performance metrics (AUC = 0.67) indicate that the identified multi-omics signature can be translated into a reliable diagnostic panel for early-stage detection. However, further biological relations need to be studied.

REFERENCES

[1] Rohart F., Gautier, B, Singh, A and Lê Cao, K. A. mixOmics: an R package for 'omics feature selection and multiple data integration. PLOS 2-17.

[2] Wu T, Hu E, Xu S, Chen M, Guo P, Dai Z, Feng T, Zhou L, Tang W, Zhan L, Fu x, Liu S, Bo X, Yu G (2021). "clusterProfiler 4.0: A universal enrichment tool for interpreting omics data." The Innovation, 2(3), 100141. doi:10.1016/j.xinn.2021.100141.

[3] Pang, Z., Lu, Y., Zhou, G., Hui, F., Xu, L., Viau, C., Spigelman, A., MacDonald, P., Wishart, D., Li, S., and Xia, J. (2024) MetaboAnalyst 6.0: towards a unified platform for metabolomics data processing, analysis and interpretation Nucleic Acids Research (doi: 10.1093/nar/gkae253)

RESULTS

DIABLO ANALYSIS

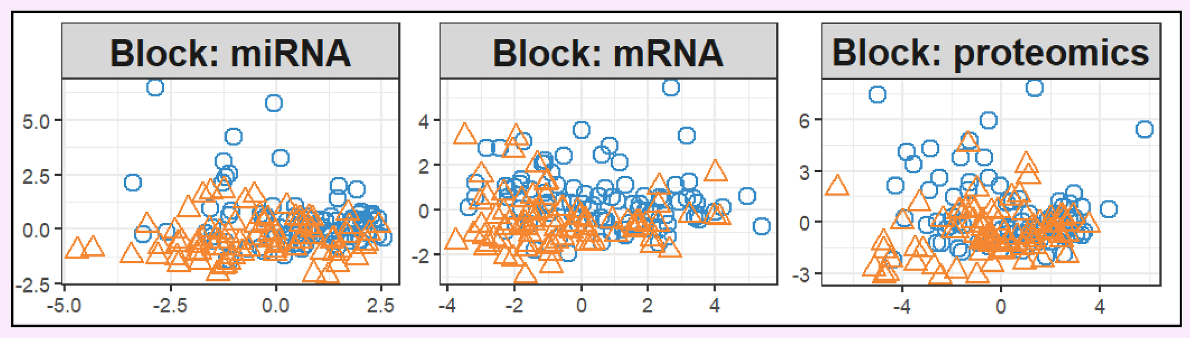


Fig 1: Sample plot from multiblock sPLS-DA illustrating the separation of tumor vs. non-tumor groups within each omics data type on the first two components.

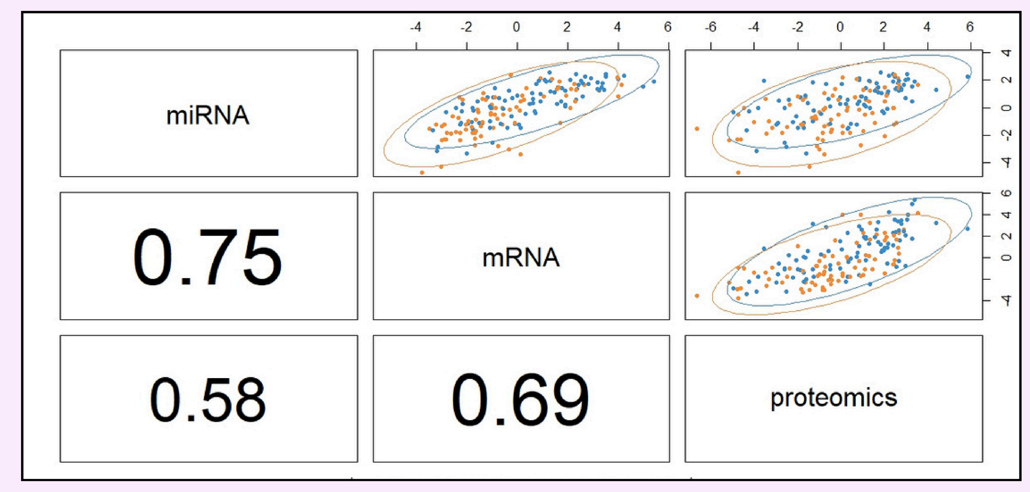


Fig 2: Diagnostic plot from multiblock sPLS-DA (ncomp = 2) showing strong inter-block correlations.

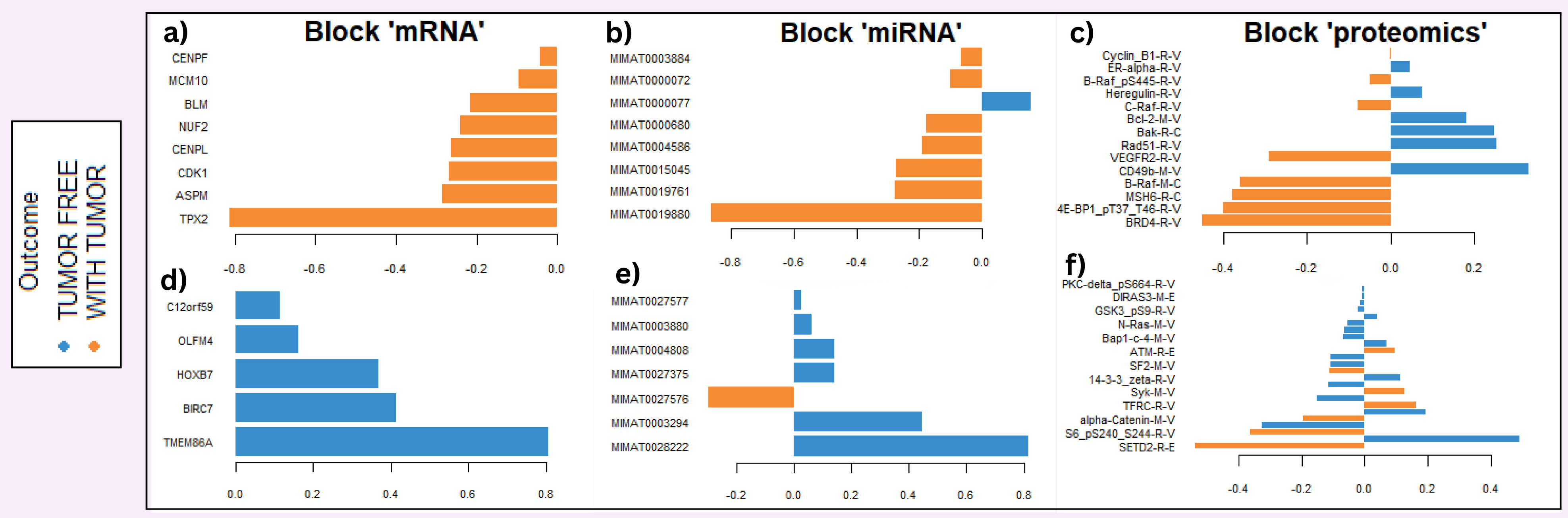


Fig 3: Loading plots identifying the key molecular features from each omics block contributing to (a-c) Component 1, (d-f) Component 2.

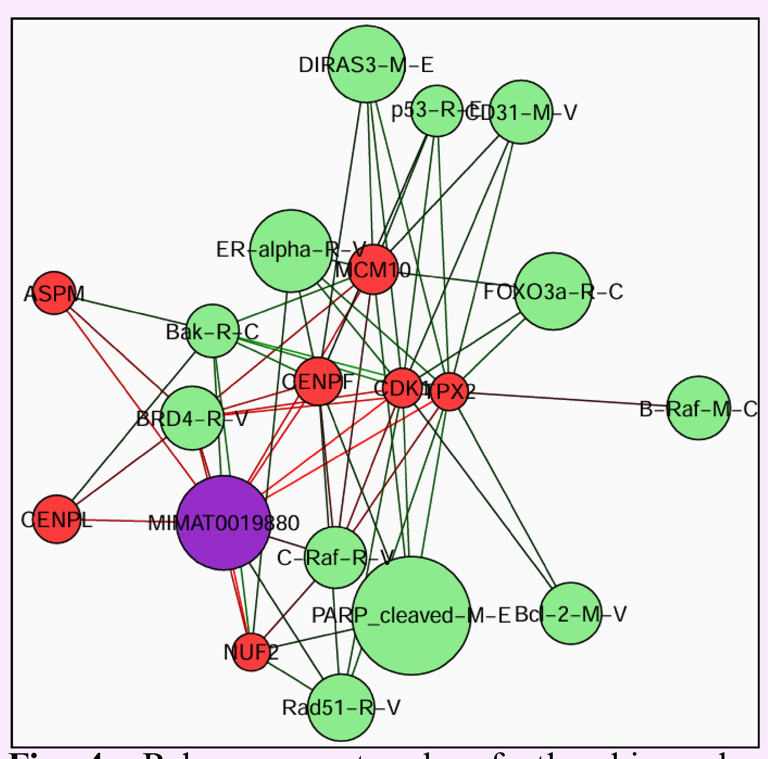


Fig 4: Relevance network of the biomarker signature, visualizing intermolecular correlations with a cutoff of r > 0.75.

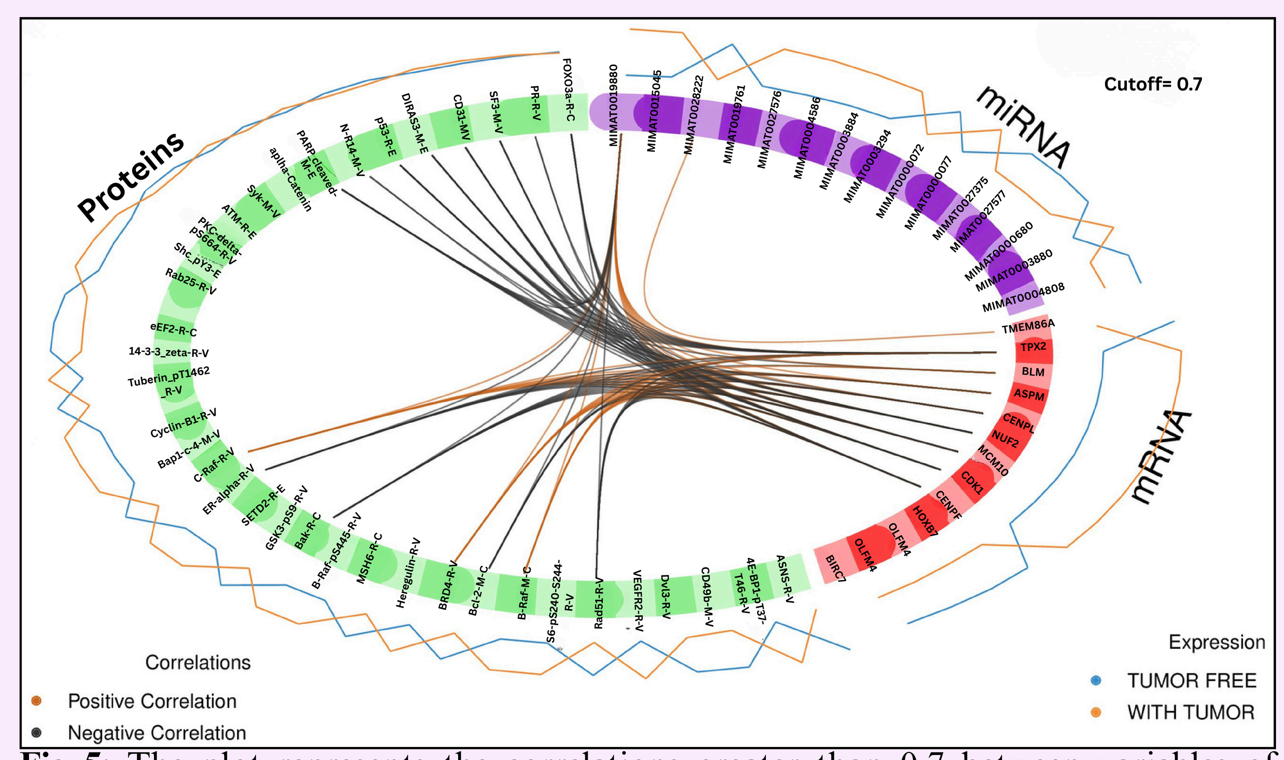


Fig 5: The plot represents the correlations greater than 0.7 between variables of different blocks.

PATHWAY ANALYSIS

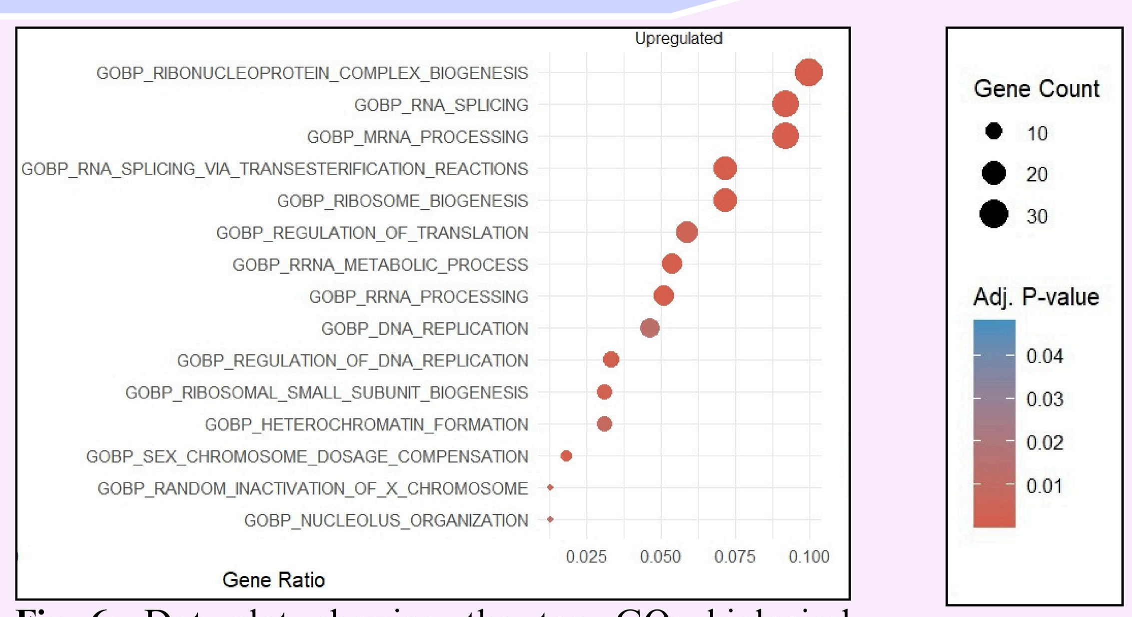


Fig 6: Dot plot showing the top GO: biological pathways enriched in upregulated genes from Principal Components 1 and 2

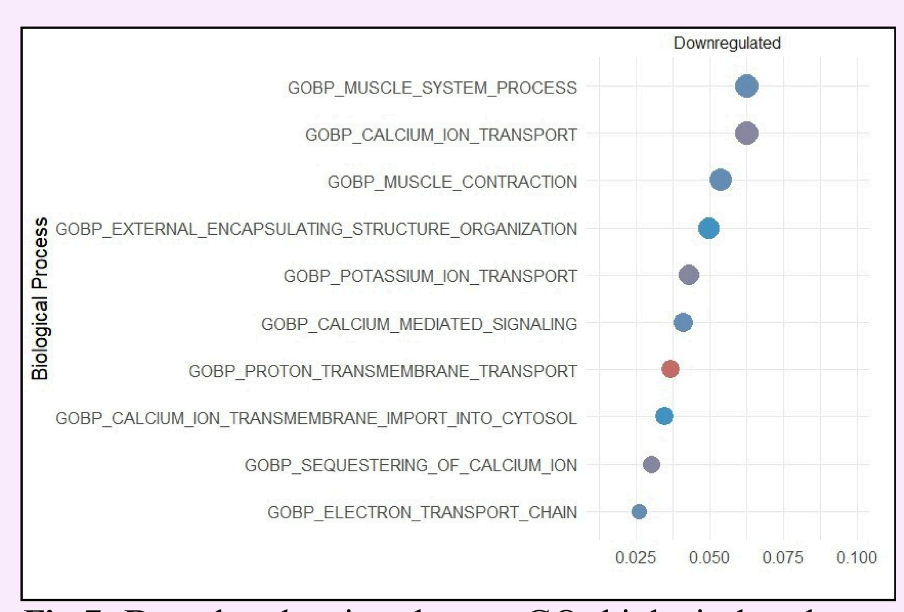


Fig 7: Dot plot showing the top GO: biological pathways enriched in downregulated genes from Principal Components 1 and 2

PERFORMANCE

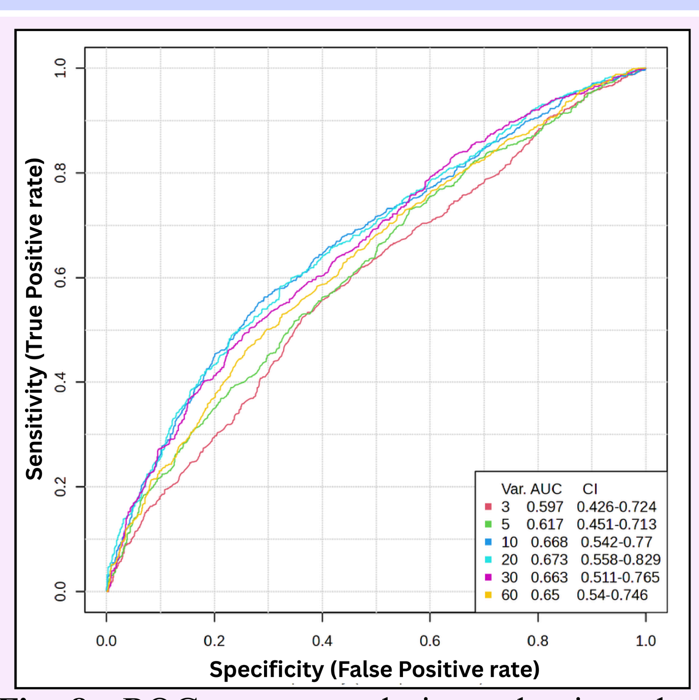


Fig 8: ROC curve analysis evaluating the performance of the classification model for predicting tumor status. Model 4 showed best performance (AUC: 0.67, Var: 20)

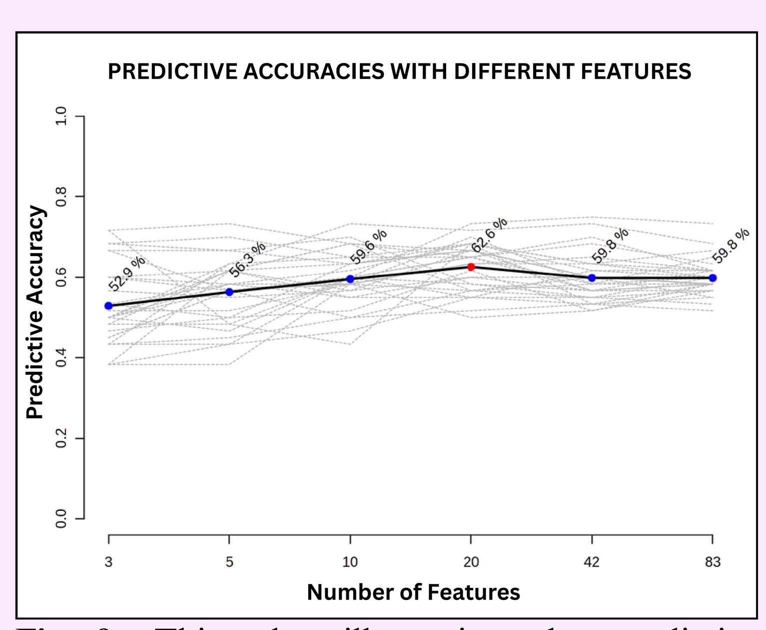


Fig 9: This plot illustrating the predictive accuracy of the classification model, assessed using different cross-validation methods. Best accuracy when no. of features is 20 (62.6%).

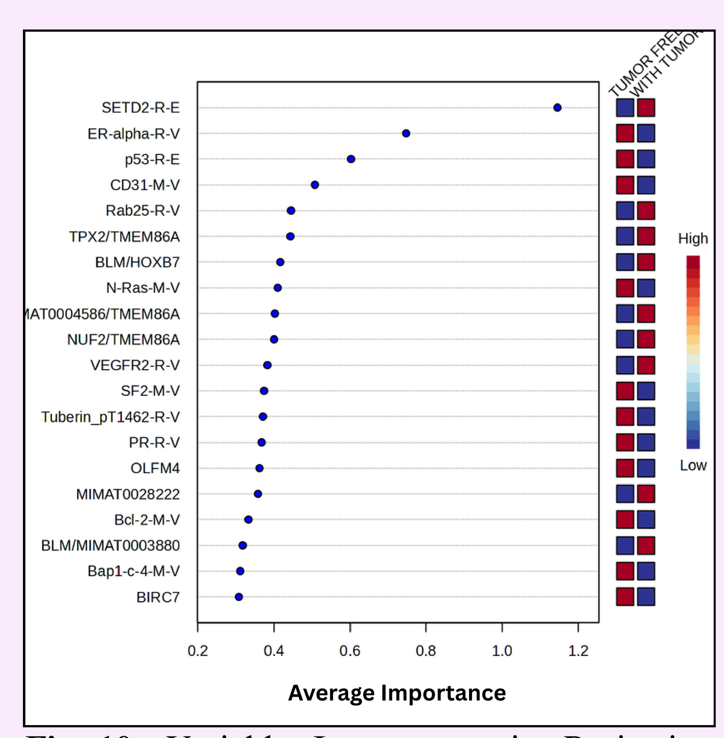


Fig 10: Variable Importance in Projection (VIP) scores identifying the top features for discriminating between sample groups to determine potential biomarkers.